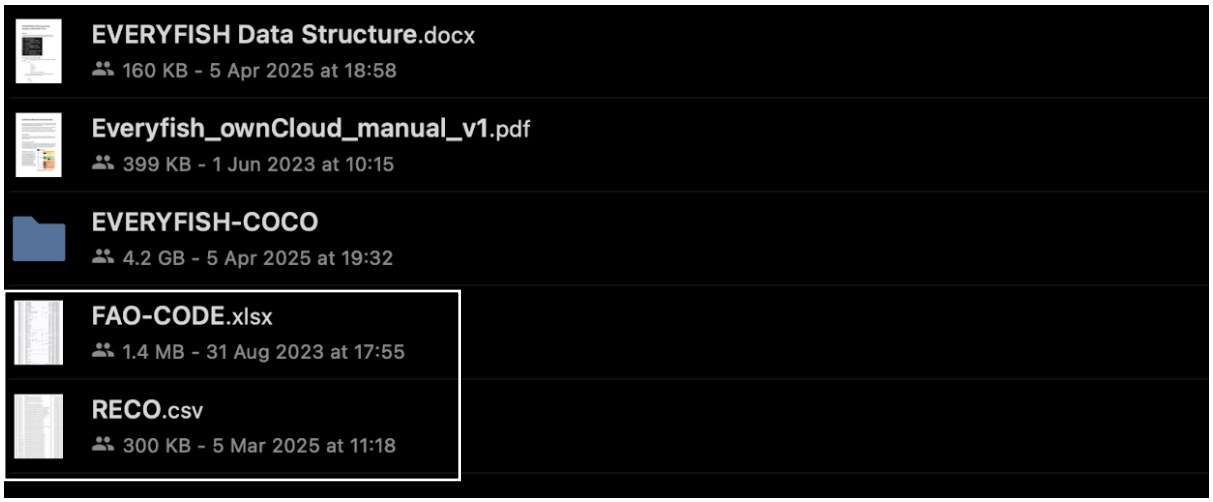
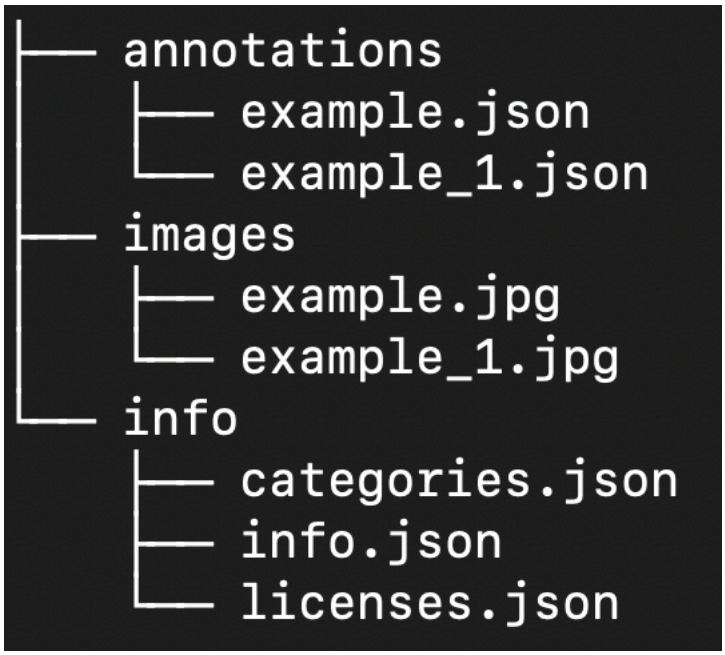
EVERYFISH COCO-based Data Structure (08/04/2025 V0.3)

Metadata documents: FAO-CODE and RECO are shared separately with this file.



General:

For each product: images, annotations and dataset meta data should be stored like the example below. Images and their corresponding annotations must have the same name.



On dataset level (“info” folder):

info.json: a json file containing year, version, description, contributor, url, contributor, date_create

‘info’: {

‘year’: str

‘version’: str

‘description’: str

‘contributor’: str

‘url’: str

‘date_created’: datetime (dd/mm/yyyy)

licenses.json: a json file containing a list of licenses applied to all images in the dataset. Each element in the list is a dictionary:

{

‘id’: int,

‘name’: str,

‘url’: str

}

categories.json: a json containing list of species names in the dataset, please include the corresponding FAO code for each species. Each element in the list is a dictionary:

{

‘id’: int

‘name’: str (Latin name or human-readable name)

‘FAO’: str (must be listed in FAO spreadsheet)

}

metadata.json: a json containing basic information of vessels/ trawls. Each element in the list is a dictionary for each image:

An example of using RECO to fill those fields

tblCodeID	Code	Description
247737	DIV_CEP_0_0_0	Diving, Cephalopods, see the code reg. mesh size and selectivity device
247738	DIV_DES_0_0_0	Diving, Benthic species, see the code reg. mesh size and selectivity device

{

“image_name”: str

“gear_type”: str

“mesh_selectivity_device”: str

“targeted_fish”: str

“fish_region”: str (the acronym of the RFMO (Regional Fisheries Management Organization) ICES, ICCAT, IATTC, IOTC and an additional one with the name of the bigger divisions of each RFMO.

For instance ICES-VII (Celtic Swa) or ICES-VIII (Bay of Biscay) for having a rough idea of where the photo has been recorded)

}

On image level (“annotations” folder):

One json per image with the same name of the image and ‘json’ as a file extension. The json file in the annotation folder should contain the following information:

“image”: {

‘license’: int (This is an ID from the “licenses.json”)

‘file_name’: str

‘width’: int

‘height’: int

‘date_captured’: datetime (dd/mm/yyyy)

‘image_id’: int

‘annotations_type’: [detection, segmentation, weight_estimation, length_estimation]

}

Additionally, in the same json file, annotations at object level should be stored. A list of object annotations must be included, one per fish instance/object. For each instance, its corresponding annotation stored like below.

“annotation”: [

{

‘id’: int (object ID)

‘image_id’: int (image ID)

‘category_id’: int (respective id from the info.json file)

‘bbox’: [x, y, width, height] (x and y coordinates from the top left image corner)

‘segmentation’: polygon (a list of points)

‘area’: float (area of the polygon in pixels)

‘fish_length’: float (in cm)

‘fish_weight’: float (in kg)

‘occlusion_type’: int [0(fully visible), 1(partially visible), 2(immeasurable)]

‘annotator_type’: [‘human’, ‘machine’ or ‘unknown’]

}

]